

## 정리. 자주쓰는 곱셈공식

$$\textcircled{1} (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$\textcircled{2} (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$\textcircled{3} a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$\textcircled{4} a^3 \pm b^3 = (a \pm b)^3 \mp 3ab(a \pm b)$$

$$\textcircled{5} a^3 + b^3 + c^3 = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) + 3abc$$

$$\textcircled{6} a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$$